

SURAJ SURVE

Physical AI · Robotics Engineer · Embodied AI · Machine Learning · GATE DA Qualified
surajsurve02022@gmail.com · +91 9766966511 · Pune, India · [LinkedIn](#) · [GitHub](#) · [GitLab](#)

3+ yrs Physical AI & Robotics — built **MANAS humanoid robot** (HRI, ROS2, Qwen 4B, Gazebo) and **6-DOF arm** with NL Brain Node + Digital Twin at M CAD Solutions. Strong in ROS2 · MoveIt2 · Deep RL · LLMs · Computer Vision · Sim-to-Real. Mentor Recognition Award R.I.D.E. 2026 (MIT-WPU). GATE DA Qualified. B.Tech AI & DS, MMCOE Pune (Grade A, 2023–27).

TECHNICAL SKILLS

Robotics & Middleware: ROS2 · ros2_control · MoveIt2 · Nav2 · TF2 · URDF/Xacro · DDS · Joint Control · Kinematics (FK/IK) · Motion Planning · Trajectory Optimization · OMPL · SLAM · Path Planning · EtherCAT · CAN · Motor Control

NVIDIA Ecosystem: Isaac Sim · Isaac Lab · Isaac ROS · Isaac Gym · Isaac GR00T · Omniverse · OpenUSD/USD · Cosmos · Jetson · JetPack · CUDA · TensorRT

Simulation: Gazebo · MuJoCo · Microsoft AirSim · Domain Randomization · Synthetic Data · Sim-to-Real Transfer · Physics-in-the-Loop · Digital Twin · Scene Generation · Hardware-in-the-Loop

Robot Learning / Physical AI: Imitation Learning · Learning from Demonstrations · Diffusion Policy · VLA (Vision-Language-Action) · VLM · Policy Learning · Teleoperation · Trajectory Collection · Dataset Curation · Agentic Task Execution · Embodied AI · Robotics Foundation Models

Reinforcement Learning: PPO · SAC · TD3 · DDPG · Double DQN (N-step) · HER · Curriculum Learning · Curiosity-Driven Exploration · Reward Engineering · Reward Shaping · Multi-agent RL · OpenAI Gym · Policy Optimization

Perception & Computer Vision: YOLO (v8/v11/v12/World/Nano) · OpenCV · PCL · Visual SLAM · Visual Odometry · VIO · EKF · Sensor Fusion (LiDAR/IMU/Camera/RGB-D) · Depth Estimation · Pose Estimation · Object Detection · Tracking · Segmentation · 3D Point Clouds

Deep Learning: PyTorch · TensorFlow/Keras · LSTM · Attention · ONNX · TensorRT · Model Quantization · HuggingFace Transformers · Distributed Training · CUDA · TensorBoard · JAX

LLMs & Conversational AI: LLM Fine-tuning · RAG Pipelines · Qwen · Ollama · Gemini 2.5 Flash · Indic STT · TTS · NLP · HRI · ChromaDB · Pinecone · Vector Embeddings

Cloud, MLOps & Systems: Docker · AWS IoT Core · GitLab CI/CD · MLOps · Edge AI · Linux · Embedded Systems · Raspberry Pi · Jetson · Git · Bash · TCP/UDP · Real-time Control

Languages: Python · C++ · C · SQL · Dart · TypeScript · Bash

Full-Stack & Mobile: FastAPI · Flask · Flutter · React · TypeScript · Node.js · PostgreSQL/Supabase (RLS) · Firebase · REST APIs

EXPERIENCE

Physical AI Engineer · [M CAD Solutions \(Manoj Potdar\)](#) · [Internship](#)Feb 2023 – Present [LINK](#)

- ▶ MANAS humanoid: ROS2 stack, Gazebo/Isaac Sim Digital Twin, YOLO+OpenCV perception, agentic task execution; real-time HRI pipeline — Indic STT → Qwen 4B LLM → TTS; iterating response latency, naturalness, regional Indian language support.
- ▶ 6-DOF robotic arm: Brain Node (Ollama LLM + YOLO World → MoveIt2 trajectory → physical execution); bidirectional Digital Twin; kinesthetic teach-by-demonstration; servo predictive maintenance; Flutter mobile app.
- ▶ Trainer at ROBONOVA 2026 (Persistent Foundation, Persistent Systems Pune, July 2026) — ROS2, URDF/Xacro, TF2, RViz2, Digital Twin bootcamp for Kiran Scholar engineers; demonstrated MANAS AI architecture live.
- ▶ Architecting AWS IoT Core migration for cloud-connected multi-robot fleet monitoring and OTA model updates.

AI Engineer · [The GrowBox Company](#) · [FMCIII Pune](#) · [Internship](#)Dec 2025 – May 2026 [LINK](#)

- ▶ Gemini 2.5 Flash Vision + PlantNet AI inference pipelines; 429 diagnoses · 382 live users · 39K+ LOC Flutter app · 15-table Supabase PostgreSQL/RLS schema · 4-stage GitLab CI/CD + Fastlane Play Store deployment.

Artificial Intelligence Intern · [Infosys Springboard](#) · [Remote](#)Dec 2025 – Apr 2026 [LINK](#)

- ▶ LSTM Smart Energy Forecasting — 95% prediction accuracy on 6-month IoT sensor data; time-series feature engineering pipeline; production Flask dashboard.

AI Engineer · [Robocon Team Rudra](#) · [On-site](#)Nov 2025 – Jan 2026 [LINK](#)

- ▶ Real-time YOLO object detection + AI decision-making for autonomous Robocon competition task execution.

Software Engineer · [GirlScript Summer of Code](#) · [Open Source](#) · [Remote](#)Oct 2024 [LINK](#)

- ▶ Open-source contributions: feature additions and bug fixes in collaborative review-driven engineering.
-

Vice President · [NEURA MMCOE](#) · [AI/ML Student Club](#) Jul 2024 – Jul 2025 [LINK](#)

- Led 100+ member AI/ML club; organised DL, CV, Robotics workshops, hackathons, speaker sessions.

PROJECTS

MANAS — Humanoid Robot · Physical AI & Conversational HRI [LINK](#) [M CAD Solutions](#)

Stack: ROS2 · ros2_control · Gazebo · Isaac Sim · URDF/Xacro · Qwen 4B · Indic STT · TTS · YOLO · OpenCV · C++ · Python

- Full-body humanoid (head, torso, dual arms/hands, legs) built from scratch — 6 months; ROS2 software stack, Gazebo + Isaac Sim Digital Twin, YOLO+OpenCV perception, agentic AI task execution.
- HRI pipeline: Indic STT → Qwen 4B LLM → TTS; target deployments: hotels, universities, showrooms, exhibitions; expanding to regional Indian languages.

6-DOF AI Robotic Arm + Digital Twin + NL Control [LINK](#) [M CAD Solutions](#)

Stack: ROS2 · MoveIt2 · Gazebo · URDF · YOLO World/Nano · Ollama LLM · AWS IoT Core · Raspberry Pi · C++ · Python

- Brain Node: NL → Ollama LLM → YOLO World detection → MoveIt2 collision-free trajectory → execution; bidirectional Digital Twin sync.
- Kinesthetic teach-by-demonstration, servo predictive maintenance, dual-camera 3D mapping, Flutter mobile app, AWS IoT fleet.

Autonomous Drone Navigation — Deep Reinforcement Learning [LINK](#) [Solo](#)

Stack: PyTorch · Microsoft AirSim · PPO · Double DQN · N-step · HER · Curriculum Learning · OpenCV · CUDA · TensorBoard

- Double DQN (N-step + HER); attention-based sensor fusion (9 LiDAR + 5 cameras → 120-dim state); 10,000+ episode convergence; curriculum learning (11 flight skills); curiosity-driven exploration.

AIVA — RAG-Based AI Voice Assistant + Agentic Execution [LINK](#) [Solo](#)

Stack: Python · Ollama (DeepSeek) · ChromaDB · Flask · Docker Compose · Tesseract OCR · STT/TTS

- RAG pipeline + agentic script execution (Python/Bash/PowerShell) with security sandboxing; Docker multi-service; 10+ format OCR ingestion; persistent MongoDB history.

GrowBox — Production AI Smart Gardening App [LINK](#) [Solo](#) · [1k+ users](#) · [13k+ AI diagnoses](#)

Stack: Flutter · Supabase · Gemini 2.5 Flash Vision · PlantNet · Next.js 16 · GitLab CI/CD · Fastlane

- 39K+ LOC; Gemini Vision + PlantNet diagnosis; 27-badge gamification; Paystack + AdMob; RLS on all 15 DB tables; 4-stage CI/CD + automated Play Store deployment.

RECOGNITION & ACHIEVEMENTS

 **Mentor Recognition Award — R.I.D.E. 2026** — MIT World Peace University, Pune · Mentored students on ROS2, Robotics, Digital Twin & Physical AI alongside Manoj Potdar (M CAD Solutions) [LINK](#)

 **Invited Trainer — ROBONOVA 2026** — Persistent Foundation Kiran Scholars · Persistent Systems Pune · July 2026 · ROS2 & Digital Twin 3-day bootcamp; live MANAS humanoid demo [LINK](#)

 **GATE DA Qualified** — Graduate Aptitude Test in Engineering — Data Science & AI · National-level exam

EDUCATION

B.Tech — Artificial Intelligence & Data Science 2023 – 2027

Marathwada Mitra Mandal's College of Engineering (MMCOE), Pune | Grade: A

Activities: VP — NEURA MMCOE · AI Engineer — Robocon Team Rudra · Cycling

CBSE HSC — Dhruv Global School, Sangamner 2023 | 80%

CBSE SSC — Vasundhara Academy, Akole 2021 | 95%

CERTIFICATIONS — 21 TOTAL (HIGHLIGHTS) [LINK](#)

MIT ML with Python (Pursuing) · **Harvard** CS50 Python · **IIT Madras** Python with Advanced AI (GUVI/Skill India Digital) · **GDSC** ML Bootcamp

Infosys Deep Learning · Generative Models · OpenAI GPT · Computer Vision 101 · Intro to Deep Learning · AI Primer · AI Builders Lab Bootcamp

· Agile Scrum in Practice